

Exact AI Retrieval **Beyond CUDA.**

A Generation-5 information-geometry substrate for smaller, faster, deterministic, hardware-independent similarity search.

[PUBLIC-SAFE INVESTOR DECK](#)

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The pitch: one encoded corpus. Any hardware. Same answers.

AI retrieval is hitting three hard walls.

RAG, semantic search, recommendation and agent memory are becoming infrastructure - but the current stack is not built for the next cost curve.

Memory

Exact search at 10M+ is dominated by bytes read, not math.

Energy

At hyperscale, per-query retrieval energy becomes an OpEx line item.

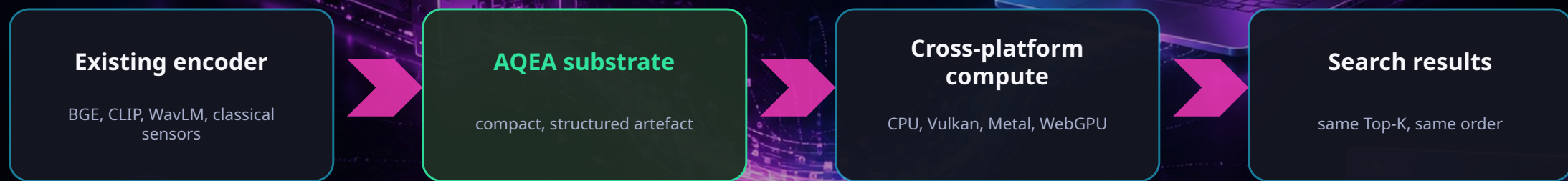
Lock-in

Production GPU retrieval is still largely tied to CUDA/NVIDIA.

The market wants exactness, lower cost and hardware choice - simultaneously.

AQEA changes the representation, not just the index.

Instead of optimizing around raw float vectors, AQEA projects encoder outputs into a structured, compact, deterministic substrate.



AQEA is a substrate layer: it can sit behind RAG, semantic search, code search, edge classification and multimodal retrieval without replacing the upstream encoder.

The proof: exact retrieval with a smaller footprint.

Production measurements against Float GPU baselines on msmarco-passage, H100 PCIe single-query mode.

1.85x

faster p50 at 10M vs Float GPU baseline

23x

smaller 10M index storage footprint

100%

Top-K bit-identity to brute-force reference

1.73x

lower energy/query at 1M steady state

4/4

GPU backends bit-identical

This is the uncomfortable part for incumbents: speed and compression are not bought by losing recall.

Why investors should care now.

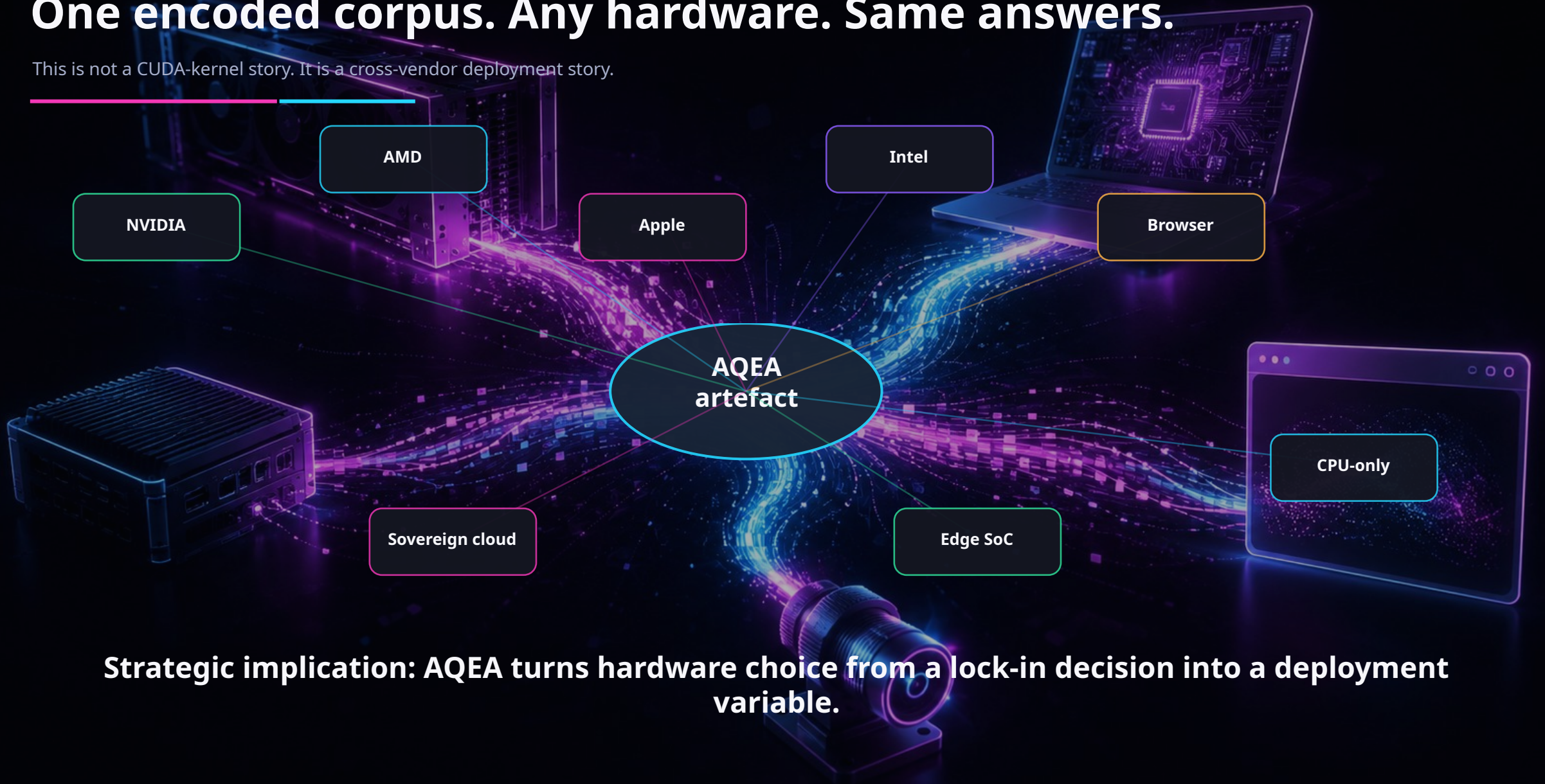
The next retrieval platform will be decided at the infrastructure layer, not the UI layer.

- RAG and agent memory make retrieval a load-bearing system, not a feature.
- Approximate search is unacceptable in legal, medical, code-security and audit-heavy workflows.
- Sovereign cloud, browser AI and edge OEMs need a path that is not CUDA-only.
- Energy and index storage become board-level economics at billion-query/day scale.

AQEA is positioned at the convergence: exactness + compression + determinism + hardware freedom.

One encoded corpus. Any hardware. Same answers.

This is not a CUDA-kernel story. It is a cross-vendor deployment story.



Strategic implication: AQEA turns hardware choice from a lock-in decision into a deployment variable.

The second wedge: edge sensing without AI accelerators.

For industrial, robotics and IoT workloads, the business case is simple: remove the dedicated accelerator from similarity-based recognition.

13/13

validated domains above floor

174%

best sensor-domain recall vs Float baseline

9.4x

max measured lower energy/query

\$20-80

AI-chip BOM removed per device class

Industrial impact: predictive maintenance, robot fleet health, wearable anomaly detection, on-device audio and regulated audit trails.

The product surface is simple.

Keep the mechanism protected. Sell the effect, the SDK and the reproducible benchmark.

Encode

Convert encoder output into compact deterministic artefacts.

Index

Load a corpus once across cloud, browser or edge targets.

Query

Return Top-K with stable ordering and reproducible results.

Verify

Hash manifests and cross-platform identity for audit paths.

Decode

Optional audit / general / retrieval operating modes.

Integration principle: bring your own encoder. AQEA handles the representation, search path and determinism layer.

Four beachheads, one substrate.

Different buyers, same core pain: retrieval must become cheaper, deterministic and less hardware-constrained.

Hyperscale RAG

Lower index footprint and retrieval energy across massive corpora.

Sovereign / multi-cloud

Exact retrieval without committing the stack to CUDA-only infrastructure.

Industrial edge

Similarity-based sensor recognition without dedicated AI accelerator BOM.

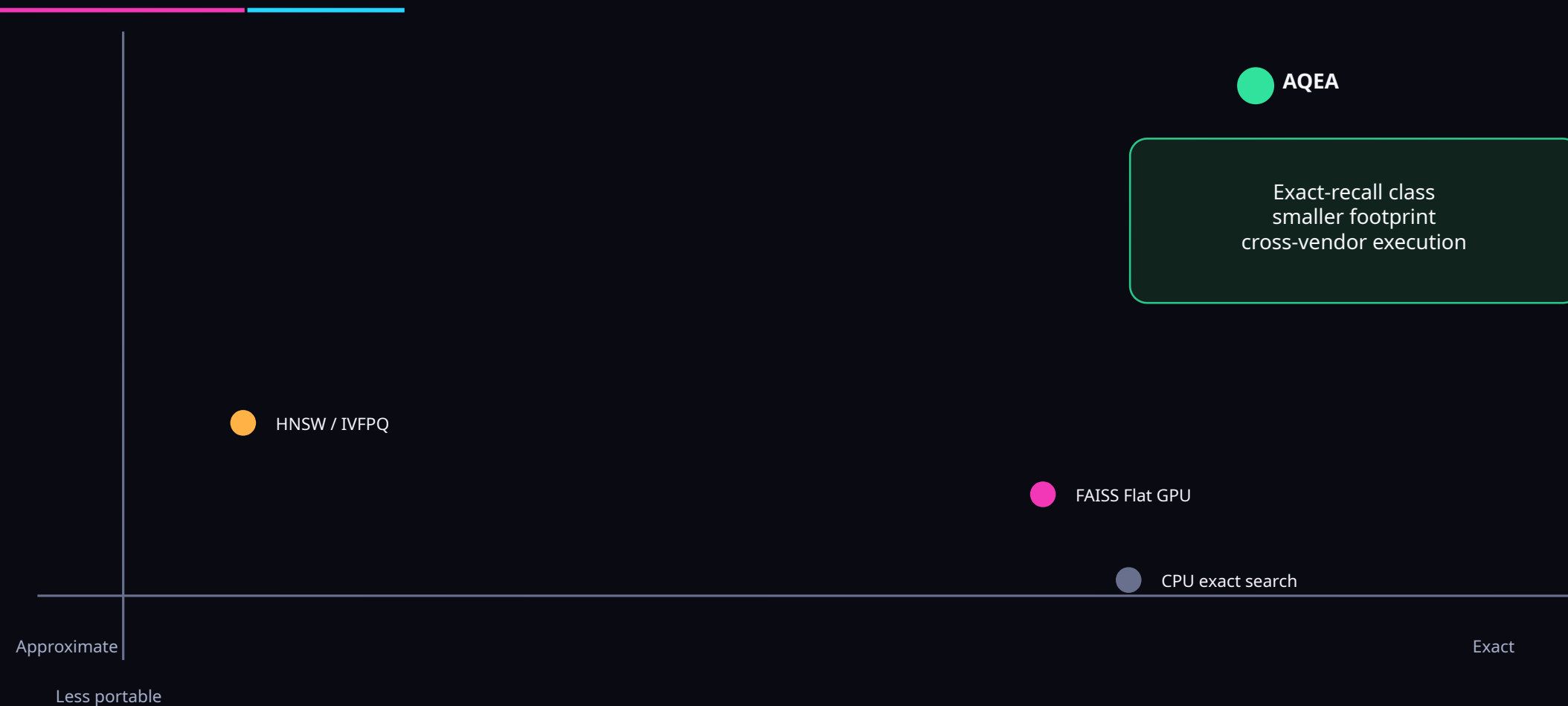
Browser / local AI

Private, local, portable retrieval in environments where CUDA does not exist.

The platform opportunity is not another vector DB. It is the representation layer underneath every vector DB.

AQEA occupies the missing quadrant.

Current choices force a trade-off: exact but expensive, or fast but approximate, or portable but weak.



Commercial motion: engineering-led, partner-first.

The fastest path to trust is to run on the partner's own workload and hardware.

1

NDA benchmark

4-6 weeks: partner corpus, partner hardware, reproducible report.

2

Shadow pilot

8-12 weeks: non-production integration against existing stack.

3

Strategic license

SDK, deployment rights, hardware tuning or co-development.

Revenue paths: evaluation fees -> pilot services -> annual SDK / runtime license -> co-development / strategic exclusivity by vertical.

Engagement options match the public-safe whitepaper partner ask: NDA evaluation, integration pilot, co-development engagement.

We are looking for the partners who cannot afford to ignore this.

For DTM / investors / strategic partners

Capital

Fund the partner proof engine, SDK hardening and GTM execution.

Strategic pilots

Run AQEA against high-value retrieval and edge workloads.

Hardware allies

Validate the same artefact across non-CUDA deployment targets.

AQEA: exact retrieval beyond CUDA.

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Appendix: evidence snapshot.

Public-safe benchmark claims used in this deck.

Cloud retrieval

10M p50: 6.27 ms AQEA vs 11.60 ms Float GPU; 1.55 GB vs 36.0 GB index.

Exactness

Top-K set, order and distance values match brute-force reference on the encoded corpus.

Cross-platform

Verified across NVIDIA Vulkan, Apple Metal and CPU reference; Edge matrix includes WebGPU.

Cross-domain

13/13 domain x encoder combinations above floor; multiple transformer and classical encoders.

Edge economics

Similarity recognition can run on standard CPU / shader APIs without dedicated AI accelerator.

Note: this deck intentionally excludes internal substrate construction, protected parameters, claim architecture and implementation details.